COURSE PROJECT SPECIFICATION

Small-to-Medium Enterprise Network Design

Course Name	Introduction to Networking (Cisco ITN) – NWC204
Team Structure	5 Students per Team
Project Duration	8 Weeks (Week 4 Checkpoint Review + Week 8 Final Defense)
Final Deliverables	1. Cisco Packet Tracer File (.pkt) 2. Technical Project Report (.docx / .pdf) 3. Short Demonstration & Oral Examination (15 Minutes)

1. Project Overview & Architecture Constraints

Teams are required to select one business model from the approved list and design a complete, multi-site network infrastructure that satisfies all operational, service, and administrative requirements.

STRICT TECHNICAL CONSTRAINTS:

• Do NOT use Virtual Local Area Networks (VLANs).
• Do NOT use Layer 3 Switches (SWL3).
• Network segregation between departments and locations must be accomplished strictly using dedicated physical switch topologies connected to distinct Router interfaces.

2. Business Scale & Requirements

- **Total Workforce:** > 200 employees represented logically in the IP addressing scheme (VLSM). In Cisco Packet Tracer, deploy 2–3 representative workstations/devices per department.
- **Departmental Structure:** > 3 distinct operational departments (e.g., Administration, Operations, IT/Servers, Sales).
- **Physical Locations:** > 1 site (Main Head Office + at least 1 Branch/Satellite Office).

3. Core Requirements (Base Grade)

A. Network Connectivity

• LAN: Physical switch-based local area networks per department/site.
• WAN: Point-to-Point WAN connections between Head Office and Branch Office (e.g., Serial PPP/HDLC links or direct Router-to-Router IP connections).
• Wireless (Wi-Fi): Access Points connected to local switches providing Wi-Fi access for mobile workforce and guests.

B. Core Network Services

- DHCP Server: Dynamic IP configuration for end-user hosts across subnets.
- DNS Server: Internal domain name resolution for organizational resources.
- HTTPS Server: Secure internal web server hosting company intranet portal.
- Email Server: Basic SMTP/POP3 mail service for internal communication.

C. Baseline Security Requirements

- Password Length & Complexity: Minimum password length configured via `security passwords min-length 10`. Strong `enable secret` passwords.
- Service Security & Encryption: Global encryption enabled (`service password-encryption`). Insecure Telnet disabled; secure SSH v2 enforced on VTY lines.
- Unused Ports & Services: Unused interfaces manually shut down (`shutdown`). Unnecessary router services disabled (e.g., `no cdp run`).
- Session & Brute-Force Controls: Executive timeout set (`exec-timeout 5 0`). Login block policies enabled (`login block-for 120 attempts 3 within 30`).
- Banners: Legal warning banner (`banner motd`) containing legal disclaimers and team identification.

4. Advanced Features (Extra Credit Tiers)

Completing Section 3 (Core Requirements) alone caps a team's grade at 7-8/10. Students who successfully implement the following advanced configuration modules are eligible for higher grade tiers (9-10/10). Note: OSPF and ACL go beyond the current syllabus (only basic static routing has been introduced in class) — teams pursuing these modules should expect to self-study the relevant Cisco documentation:

Advanced Module	Technical Scope & Requirements
Dynamic Routing Protocols	Implement OSPFv2 across multi-site routers with proper area configurations instead of static routes.
Access Control Lists (ACLs)	Configure Standard or Extended ACLs to enforce departmental isolation, restrict server farm access, or grant internet-only access to guests.
AAA Security Architecture	Deploy AAA authentication using RADIUS or TACACS+ servers for centralized device management and network user access.
IPv6 Dual-Stack Deployment	Configure full IPv6 dual-stack addressing across LAN and WAN interfaces alongside IPv4.

5. Project Governance & Submission Standards

To ensure consistency, traceability, and authentic project execution across all teams, the following administrative and technical standards must be strictly followed:

1. Assigned Network Parameters: Each team must configure their topology using the unique base IP addressing block provided by the instructor during Week 2.

2. Device Identification & Ownership: All active devices must reflect the team designation in their hostnames (e.g., `R1-Group05-HQ`), local domain setting (`group[XX].[company].com`), and login banners (`banner motd`).

3. Environment & User Profile Integrity: Students must complete their team information under `Options -> User Profile` in Cisco Packet Tracer prior to building the network topology.

4. Mid-Term Checkpoint: A mandatory progress review will occur at the end of Week 4. Teams will present their core LAN/WAN connectivity and IP addressing tables to receive feedback before moving to advanced configurations.

5. Interactive Final Defense: The 15-minute final demonstration includes brief interactive Q&A and verification steps where team members demonstrate their direct understanding of the submitted configuration.